

Week 5 — Demand-Signal & Customer-Behavior Methodology

inGen Dynamics service-robotics portfolio · search interest + news cadence/sentiment + voice-of-customer, combined into one comparable demand-signal index across five verticals.

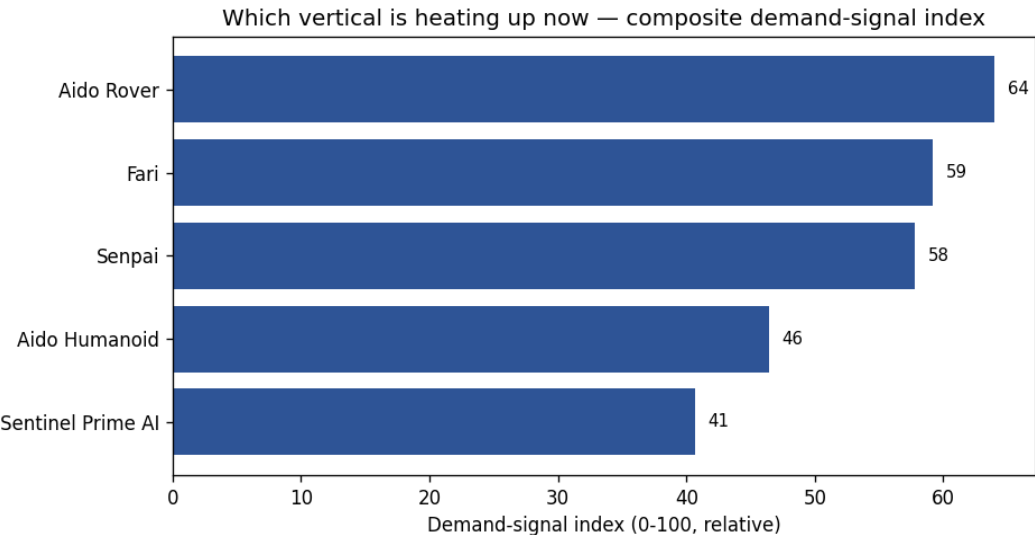
1. Signal families & sources

Search interest — Google Trends (US) monthly interest for curated keywords per vertical (pytrends). **News cadence & sentiment** — GDELT DOC 2.0 API: monthly article volume (TimelineVolRaw) and average tone (TimelineTone). **Voice-of-customer** — review/comment text per vertical, clustered with TF-IDF + KMeans into themes; the most negative review per theme is surfaced as the pain point.

2. Demand-signal index (documented weights)

Each component is normalised 0-100 across verticals, then combined: search momentum 40% (last-12-mo vs prior-12-mo), news momentum 35% (last-6-mo vs prior-6-mo volume), news sentiment 25% (mean tone, last 6 mo). The index is a **relative** read of which vertical is heating up now — not an absolute forecast.

Rank	Vertical / product	Index	Search	News vol	Sentiment
1	Aido Rover (outdoor_patrol)	64.0	72	100	0
2	Fari (eldercare)	59.2	81	47	41
3	Senpai (education)	57.8	0	94	100
4	Aido Humanoid (humanoid)	46.4	100	0	25
5	Sentinel Prime AI (indoor_security)	40.7	18	58	52



3. Sentiment model choice (justification)

Two sentiment layers are used. For **news**, GDELT's document-level **tone** metric is taken directly (a validated, corpus-wide score on a consistent scale across all 24 months and verticals), which avoids re-scoring millions of articles and keeps the signal comparable. For **voice-of-customer**, themes come from **TF-IDF + KMeans** and each theme's pain point is the review with the highest count of negation/■complaint terms from a fixed lexicon. This lexicon approach was chosen over a heavier spaCy/NLTK or transformer classifier because it is fully **deterministic and reproducible** (no model download, no random init beyond fixed-seed KMeans), transparent (the ranking rule is inspectable), and adequate for short review text; the same code accepts a spaCy/NLTK or BERTopic backend if richer sentiment is later required. Choice is documented here per the brief.

4. Reproducibility & honesty

Run mode for this build: **offline-fallback (synthetic, schema-correct)**. Real public APIs are wired in each collector; when network is unavailable the pipeline uses clearly-labelled schema-correct samples so it runs end-to-end, and the manifest (signal_manifest.jsonl) records mode per source. Set INGEST_ALLOW_NETWORK=1 to populate from live Google Trends and GDELT, and drop real reviews_<vertical>.csv files for live voice-of-customer.

Prepared by Ziheng Wang · inGen Data Analyst internship · Week 5 · 2026-06-16. Index is a relative demand read from public signals, not a forecast; validate against internal pipeline/CRM data before planning decisions.